1

How many subset of an empty set?

A) 0 B) 1

C) 2 D) other

2

$f(n)=1\cdot 2+2\cdot 3+3\cdot 4+\cdots+(n-1)n$ is

A) O(n) B) $O(n^2)$ C) $O(n^3)$ D) $n^2\log n$ E) nlog n

3

( ) If the sets A, B and C satisfy the following relations simultaneously:

$A\cup B=C;\ (A\cup C)\cap B=C;\ (A\cap C)\cup B=A$

Then $A=B=C$

4

( ) The function $f(n)=\left\lceil\frac{2n-1}{2}\right\rceil$ from **Z** to **Z**, This is one-to-one corresponding function.

5

( ) The set of complex numbers has the same cardinality as the set of real numbers

6

( ) Let *S* be the set of all finite subsets of , then *S* is countable.

7

( ) Let *S* be the set of all infinite subsets of , then *S* is uncountable.

8

( ) $\sum_{k=0}^{n} k\cdot\binom{n}{k}=O\left(2^{n\log_3 n}\right)$

9

( ) Let $f=A\to B$ and $g=B\to C$ be functions. If g°f is injective, then *f* is injective.

10

( ) Let $f = A \to B$ and $g = B \to C$ be functions. If $g \circ f$ is surjective, then f is surjective.

11

( ) The proposition $(p \to q) \wedge (q \to \neg p) \wedge (p \vee q)$ is not satisfiable.

12

( ) $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 0.5 \rfloor$

13

Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by giving an element table proof.

14

Let $P(m,n)$ be the statement "$m \mid n$," where the domain for both variables consists of all positive integers. [By "$m \mid n$", which we say as "m divides n", we mean that $n = km$ for some integer k.] Determine the truth values of each of these statements.

(a) $P(4,5)$ **(b)** $P(2,4)$ **(c)** $\forall m \forall n\, P(m,n)$
(d) $\exists m \forall n\, P(m,n)$ **(e)** $\exists n \forall m\, P(m,n)$ **(f)** $\forall n\, P(1,n)$

15

Show that $\neg p \to (q \to r)$ and $q \to (p \vee r)$ are logically equivalent using the laws of logical equivalences. Be sure to cite each law whenever used.

And give their full DNF and full CNF

16

(2 marks) A sequence is defined recursively by : $c_k = 3c_{k-1} + 1, k \geq 2$ and $c_1 = 1$.
a) Use iteration to guess an explicit formula for this sequence.
b) Now use mathematical induction to verify the correctness of your formula.

17

Use the Euclidean algorithm to find

a) gcd(1, 5). **b)** gcd(100, 101).
c) gcd(123, 277). **d)** gcd(1529, 14039).
e) gcd(1529, 14038). **f)** gcd(11111, 111111).